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Conformal selective prediction with cost aware deferral for safe clinical triage under distribution shift

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We propose a selective prediction framework for clinical triage that combines calibrated probabilistic modeling, conformal prediction, and cost-aware deferral to prioritize patient safety. The system produces set-valued predictions with finite-sample coverage control and defers low-confidence cases to clinicians when doing so reduces an explicit expected clinical cost. We construct prediction sets using split conformal prediction, a group-conditional Mondrian variant for gender-stratified coverage, and an importance-weighted variant to improve robustness under distribution shift, and we select a single deferral threshold by minimizing expected cost on a held-out calibration set. On temporally separated in-distribution and out-of-distribution test splits for early sepsis prediction, selective deferral yields a favorable risk–coverage trade-off, reducing error on retained cases by 49.6% on the in-distribution (ID) test split and 46.7% on the out-of-distribution (OOD) test split, both at 80% coverage, while achieving low expected cost on both splits with only a moderate increase under shift. Calibration remains strong with low expected calibration error on both test sets, and rule-out performance is conservative, attaining near-perfect negative predictive value at a 95% sensitivity target. Coverage stays close to the nominal 90% target in-distribution and degrades only slightly out-of-distribution, with the weighted method most robust, and the Mondrian method reduces the gender coverage gap to 1.4 percentage points. These results indicate that conformal uncertainty quantification combined with cost-aware deferral can provide transparent and safer clinical decision support that degrades gracefully under temporal distribution shift.

Keywords Selective prediction, Conformal prediction, Reject option, Calibration, Clinical decision support, Sepsis, Out-of-distribution robustness, Fairness

Clinical decision support systems[1–3] increasingly rely on supervised learning to triage high-risk patients[4,5], prioritize scarce clinical resources[6,7], and standardize care pathways[8,9]. Yet conventional model selection and reporting practices are dominated by summary discrimination metrics such as AUROC[10], which neither account for asymmetric clinical harm nor recognize that in safety-critical settings a model should sometimes abstain from prediction rather than commit to an uncertain label. This limitation is amplified by distribution shift between historical development data and prospective deployment contexts, where calibration and error characteristics can drift over time. In problems such as early sepsis detection, the cost of a false negative can far exceed the cost of a false alarm, and clinical workflows naturally incorporate escalation and deferral to human experts. A principled learning framework should therefore combine uncertainty quantification, explicit deferral, and clinically meaningful objectives rather than optimizing accuracy alone.

A long line of research studies uncertainty and abstention. Probability calibration methods[11,12], including temperature scaling, improve the agreement between predicted confidences and empirical frequencies without altering the underlying decision rule, but they do not specify when to defer or how to trade prediction against abstention. Classical reject-option learning[13,14] formalizes abstention by introducing a separate reject action with an associated cost, but practical deployments often rely on ad hoc thresholds and lack performance guarantees under shift. Selective prediction architectures similarly learn when to predict versus abstain, yet their guarantees are typically asymptotic or model-dependent, which complicates clinical assurance and auditing. In parallel, conformal prediction constructs set-valued outputs with finite-sample coverage guarantees under exchangeability, providing a distribution-free way to express uncertainty that is directly checkable from data. Mondrian conformal predictors[15,16] refine coverage within predefined subgroups, which is attractive for

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fairness-aware auditing in healthcare, while weighted conformal variants adapt to covariate shift by reweighting calibration scores. However, conformal prediction by itself does not choose an operational point, nor does it encode the relative harm of false negatives, false positives, and deferrals that drive real clinical decisions.

This paper introduces Selective Prediction in Medicine, a triage framework that integrates calibrated probabilistic modeling, conformal prediction, and a cost-aware deferral policy to prioritize patient safety. We train a base classifier for 6-hour early sepsis prediction on the PhysioNet 2019 cohort[17], calibrate its probabilities on a held-out split, and construct conformal sets using three variants: vanilla split conformal prediction, a Mondrian version stratified by gender to assess groupwise coverage, and a weighted version that targets robustness under temporal shift. We then learn a single, transparent deferral threshold by minimizing an explicit expected clinical cost on the calibration set, reflecting the triage principle that missed sepsis is far costlier than an extra review by a clinician. The resulting decision rule predicts when sufficiently confident and otherwise defers, yielding an operational risk–coverage trade-off that can be tuned to local resource constraints and harm preferences.

Our evaluation uses temporally stratified in-distribution and out-of-distribution test splits to emulate prospective deployment. The framework achieves a favorable risk–coverage profile: at 80% coverage, the error rate on retained cases is reduced by 49.6% on in-distribution data and 46.7% under temporal shift. Optimizing the deferral threshold at a single operating point yields low expected clinical cost on both splits, with a moderate relative increase under shift that is acceptable for monitored deployment. Calibration remains strong after temperature scaling, with expected calibration error below 0.03 even OOD[18], which supports trustworthy uncertainty estimates and the validity of conformal sets. At a clinically relevant sensitivity target of 95%, negative predictive value is approximately one on both splits, indicating safe rule-out behavior consistent with conservative triage. Among conformal variants, coverage on in-distribution data remains close to the nominal 90% target, and degradation OOD is limited; weighted conformal prediction is most robust to shift, while Mondrian conformal prediction narrows gender coverage disparity to near parity, aligning with fairness considerations in clinical machine learning.

The contributions of this work are threefold. First, it formulates clinical selective prediction as a cost-aware decision problem that unifies calibrated probabilities, conformal set construction, and a transparent deferral rule chosen on a calibration split. Second, it operationalizes guarantees into practice by demonstrating stable risk–coverage improvements, low expected cost, and strong calibration across temporal shift, thereby addressing deployment concerns that discrimination metrics overlook. Third, it examines subgroup coverage through Mondrian conformal prediction and robustness through weighted conformal prediction, showing that fairness and shift-awareness can be incorporated without sacrificing efficiency of the prediction sets. An open implementation using only public datasets accompanies the study to facilitate reproduction and adaptation.

The remainder of the paper is organized as follows. Section 2 reviews selective prediction, reject-option learning, calibration, conformal prediction, and their use in clinical AI. Section 3 presents the proposed triage framework, including cost modeling, conformal set construction, and deferral optimization. Section 4 describes datasets, preprocessing, models, and metrics, and reports results on in-distribution and out-of-distribution splits together with fairness and robustness analyses. Section 5 discusses practical considerations, limitations, and implications for deployment. Section 6 concludes with future directions.

Related work

Evaluation practices and clinical safety

Clinical decision support models[19,20] are often developed and reported using discrimination metrics such as AUROC and AUPRC[21]. While these summaries are useful for ranking models, they do not encode asymmetric clinical harm or the operational need to abstain in the face of uncertainty. In high-stakes tasks such as early sepsis detection, the marginal utility of sensitivity and the relative cost of false negatives versus false positives dominate the evaluation problem. Decision-theoretic[22] and cost-sensitive learning[23] frameworks propose optimizing thresholds to reflect application-specific utilities, but they are rarely coupled to principled uncertainty quantification or to explicit defer-to-human pathways in deployment. This gap motivates selective prediction regimes in which the system may decline to predict when confidence is low, thereby trading coverage for reduced risk on retained cases.

Probability calibration and uncertainty estimation

Calibration[24] aligns predicted probabilities with empirical frequencies so that confidence scores can be trusted in decision making. Modern neural networks are known to be miscalibrated, often overconfident, which has spurred practical post-hoc methods such as temperature scaling and related techniques including Platt scaling and isotonic regression. Bayesian[25] approximations like Monte Carlo dropout interpret dropout as variational inference and produce uncertainty estimates by sampling, while deep ensembles obtain uncertainty through diversity across independently trained models. These approaches improve probability quality but do not specify when to abstain or how to incorporate abstention costs. Moreover, calibration can degrade under distribution shift, which limits the reliability of confidence scores without additional safeguards.

Selective prediction and the reject option

Selective prediction[14,26], also known as classification with a reject option, extends the decision set with an abstain action that incurs a specified cost. Classical analyses characterize the optimal reject rule in terms of posterior probabilities and a reject cost, yielding a threshold on the maximum class probability. Recent deep learning methods learn joint prediction-selection mechanisms or specialized heads that estimate coverage controls, and they popularize the risk–coverage curve as a diagnostic. Although these methods formalize abstention, they typically require model-dependent training procedures or deliver asymptotic guarantees that are difficult to

audit prospectively. In clinical contexts, assurance arguments benefit from finite-sample guarantees and from transparent selection policies that can be tuned and reviewed on held-out data.

Conformal prediction and its extensions

Conformal prediction[27] constructs set-valued outputs with distribution-free, finite-sample coverage guarantees under exchangeability. Split conformal prediction calibrates a nonconformity score on a held-out calibration set to obtain a quantile threshold, then includes labels in the prediction set when their scores fall below this threshold. For structured deployment concerns, several extensions are relevant. Mondrian conformal prediction stratifies calibration and testing by predefined groups, enabling group-conditional guarantees that are attractive for fairness-aware auditing in healthcare. Weighted conformal variants reweight calibration scores to better match a shifted test distribution, improving robustness under covariate shift while retaining a clear construction based on quantiles. In classification settings, these ideas yield compact label sets that communicate uncertainty directly and can be combined with conventional point decisions when operational needs require a single action.

Fairness and distribution shift in clinical machine learning

Real-world clinical datasets[28] exhibit systematic differences across demographic groups and across time. Work on fairness in healthcare highlights how models can propagate or amplify disparities if subgroup performance is not monitored and controlled. Group-conditional calibration and coverage are therefore important auditing targets. Separately, temporal and institutional shifts degrade discrimination and calibration, motivating methods that either monitor and recalibrate post-deployment or adapt to the new distribution by importance weighting. In addition to these strategies, simple confidence-based deferral can act as a safety valve by routing difficult or distributionally novel cases to clinicians, but doing so responsibly requires a principled link between uncertainty and clinical cost.

The present work draws from these strands but combines them in a way that targets clinical safety[29,30] and auditability[31,32]. In contrast to pure calibration methods, which improve probability quality without specifying an operating policy, our approach integrates temperature scaling with conformal set construction so that uncertainty is communicated both as calibrated scores and as coverage-controlled sets. Unlike selective prediction architectures[33] that require specialized training objectives or selection heads, we adopt a transparent, post-hoc deferral rule based on the maximum posterior and choose its threshold on a held-out calibration split to minimize an explicit expected clinical cost in the spirit of classical reject-option analysis. Compared to conformal prediction alone, which guarantees marginal coverage but leaves the decision threshold unspecified, we explicitly couple conformal outputs with a cost-aware operating point that prioritizes false-negative avoidance. Finally, by instantiating Mondrian conformal prediction and a weighted variant, we address subgroup coverage balance and robustness to temporal shift within the same framework. The result is a selective predictor that preserves finite-sample guarantees where applicable, exposes a tunable and auditable deferral policy, and optimizes a clinically meaningful objective rather than a proxy discrimination metric.

Proposed method

This work introduces a selective prediction framework for clinical triage that joins calibrated probabilistic classification, conformal prediction, and a cost-aware deferral policy into a single operational pipeline. The goal is to deliver predictions when the model is sufficiently confident and to defer otherwise, while retaining distribution-free coverage guarantees for the uncertainty sets and minimizing an explicit surrogate for clinical harm. The method is designed to be simple to audit and tune: a base classifier is trained on historical data; a held-out calibration split is used both to adjust probability sharpness and to select operating thresholds; conformal prediction constructs set-valued outputs whose coverage can be monitored overall and by subgroup; and a single deferral threshold is chosen to minimize expected clinical cost. The resulting system exposes a transparent risk–coverage trade-off and a small number of interpretable knobs, which makes it amenable to prospective governance. Figure 1 shows data preprocessing, base model training, probability calibration, conformal set construction, cost-based deferral selection on the calibration split, and evaluation on in-distribution and out-of-distribution cohorts.

Method overview

Let (X, Y) denote inputs and labels, with $Y \in \{0, 1\}$ for the early sepsis task, and let f_θ be a probabilistic classifier trained on a development set $\mathcal{D}_{\text{train}}$. Given logits $z_\theta(x)$, we obtain calibrated probabilities $p_T(y \mid x)$ by a scalar temperature $T > 0$ fitted on a disjoint calibration set $\mathcal{D}_{\text{cal}}$. From these calibrated scores, we derive conformal prediction sets that communicate uncertainty as label sets with nominal coverage, using either vanilla

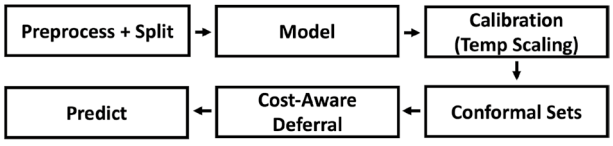


Fig. 1. Keyword-level overview aligned with the method text: preprocess and split, base model training, calibration on the calibration split, conformal set construction (vanilla, Mondrian, weighted) with coverage monitoring, cost-aware deferral threshold $\tau^\star$ optimized on the calibration split, decision (predict or defer) with set reporting.

split conformal prediction, a Mondrian variant that conditions on a protected attribute for groupwise coverage, or a weighted variant that adapts to covariate shift via importance weighting. Finally, a single deferral threshold $\tau \in (0,1)$ is selected on $\mathcal{D}_{\text{cal}}$ to minimize an explicit expected clinical cost with asymmetric penalties for false negatives, false positives, and deferrals. At deployment, a case is predicted if its certainty exceeds τ; otherwise it is routed to human review. This separates model development from operating policy selection, maintains finite-sample coverage where applicable, and encodes the safety preference that missed sepsis is substantially more harmful than an extra alarm or a deferred case.

Formalization and components

The base classifier produces logits $z_\theta(x) \in \mathbb{R}^K$ with $K = 2$ classes for the sepsis task. Temperature scaling applies a single scalar $T > 0$ and maps logits to calibrated probabilities

$$p_T(y \mid x) \;=\; \frac{\exp\{z_\theta(x)_y / T\}}{\sum_{y'} \exp\{z_\theta(x)_{y'} / T\}}, \tag{1}$$

where T is chosen to minimize the negative log-likelihood of $\mathcal{D}_{\text{cal}}$. Denote by $\hat{y}(x) = \arg\max_y p_T(y \mid x)$ the maximum a posteriori class and by $u(x) = \max_y p_T(y \mid x)$ the associated certainty score.

Split conformal prediction constructs set-valued predictions with finite-sample, distribution-free coverage under exchangeability. For binary classification we use the nonconformity score $s(x,y) = 1 - p_T(y \mid x)$. On the calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^m$, compute $a_i = s(x_i, y_i)$ and their $(1-\alpha)$-quantile with the canonical finite-sample correction

$$q_\alpha \;=\; \text{Quantile}_{\lceil (m+1)(1-\alpha) \rceil / m}\big(\{a_i\}_{i=1}^m\big). \tag{2}$$

For a new input x, the conformal prediction set is

$$\mathcal{C}_\alpha(x) \;=\; \{\, y \in \{0,1\} : s(x,y) \le q_\alpha \,\}. \tag{3}$$

Under exchangeability, $\mathbb{P}\{Y \in \mathcal{C}_\alpha(X)\} \ge 1 - \alpha$. Mondrian conformal prediction refines this construction by computing separate quantiles $q_{\alpha,g}$ within groups $g \in \mathcal{G}$ defined by a protected attribute (for example, gender), yielding group-conditional coverage $\mathbb{P}\{Y \in \mathcal{C}_{\alpha,g}(X) \mid G = g\} \ge 1 - \alpha$. Weighted conformal prediction improves robustness under covariate shift by replacing the empirical quantile with a weighted quantile. Let $w_i \propto r(x_i)$ approximate the density ratio between test and calibration covariates, for example via a logistic regression classifier that discriminates calibration versus target samples. The weighted threshold $\tilde{q}_\alpha$ is the smallest t such that

$$\sum_{i=1}^m w_i \, 1\{a_i \le t\} \;\ge\; (1-\alpha) \sum_{i=1}^m w_i. \tag{4}$$

Replacing q_α with $\tilde{q}_\alpha$ in the construction above yields prediction sets that better match the shifted target distribution.

The deferral policy is defined on the calibrated probabilities and optionally on the conformal set size. We adopt a certainty-threshold rule

$$\delta_\tau(x) \;=\; 1\{u(x) < \tau\}, \tag{5}$$

which defers when $\delta_\tau(x) = 1$ and predicts $\hat{y}(x)$ otherwise. In applications where indecision should also trigger deferral, one may augment the rule by deferring whenever $|\mathcal{C}_\alpha(x)| > 1$. We focus on the certainty-threshold rule for simplicity and transparency, and we choose τ on $\mathcal{D}_{\text{cal}}$ by minimizing an expected clinical cost. For Weighted CP, we assume access to *unlabeled* covariates from the OOD evaluation window to train a logistic discriminator against calibration covariates; the resulting probabilities are converted into density-ratio weights for the weighted quantile.

Cost model and operating point

Let $c_{\text{FN}}, c_{\text{FP}}, c_{\text{DEF}} > 0$ denote the costs of a false negative, a false positive, and a deferral, respectively, with $c_{\text{FN}} \gg c_{\text{FP}}$ to reflect the high harm of missed sepsis. For a candidate τ, define on $\mathcal{D}_{\text{cal}}$ the empirical expected cost

$$\begin{aligned} \mathcal{L}_{\text{cal}}(\tau; \alpha) = \frac{1}{m} \sum_{i=1}^m \Big[& c_{\text{DEF}} \, 1\{\delta_\tau(x_i) = 1\} \\ & + c_{\text{FN}} \, 1\{\delta_\tau(x_i) = 0,\, \hat{y}(x_i) = 0,\, y_i = 1\} \\ & + c_{\text{FP}} \, 1\{\delta_\tau(x_i) = 0,\, \hat{y}(x_i) = 1,\, y_i = 0\} \Big]. \end{aligned} \tag{6}$$

The selected operating point is

$$\tau^\star \in \arg\min_{\tau \in \mathcal{T}} \mathcal{L}_{\text{cal}}(\tau; \alpha), \tag{7}$$

for a discrete search grid $\mathcal{T}$. If α is not fixed a priori, a two-dimensional grid over (α, τ) on $\mathcal{D}_{\text{cal}}$ can be used, with the understanding that α affects set sizes and marginal coverage while τ trades coverage for risk on the retained subset. The same operating point is then evaluated on in-distribution and out-of-distribution test sets without further tuning.

The operating characteristics of the resulting rule can be summarized by the risk–coverage curve. Let $\kappa(\tau) = \mathbb{P}\{u(X) \geq \tau\}$ denote coverage and

$$R(\tau) = \mathbb{P}\{\hat{y}(X) \neq Y \mid u(X) \geq \tau\} \tag{8}$$

denote risk on non-deferred cases. Increasing τ decreases coverage and typically decreases risk on the retained subset; the cost-aware selection of $\tau^\star$ chooses a point on this curve that aligns with clinical preferences encoded in $c_{\text{FN}}, c_{\text{FP}}, c_{\text{DEF}}$. In addition, safety-critical operating points such as a target sensitivity can be reported, for example by tuning a class-specific decision threshold on $p_T(1 \mid x)$ subject to $\text{Sensitivity} \geq s_0$ and then computing negative predictive value at that point.

The complete procedure is summarized as Algorithm 1. It takes as input disjoint training, calibration, and test splits, the cost vector, and a search grid for τ (and optionally α). It returns a calibrated classifier, conformal thresholds for vanilla, Mondrian, or weighted construction, and the cost-minimizing deferral threshold. The same construction can be instantiated with groupwise thresholds for fairness auditing or with weighted quantiles for shift-robustness, without modifying the optimization of τ.

Require: $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{id}}, \mathcal{D}_{\text{ood}}$; costs $(c_{\text{FN}}, c_{\text{FP}}, c_{\text{DEF}})$; grids $\mathcal{A}$ for α and $\mathcal{T}$ for τ; optional groups $g(x)$; optional weights via $w(x)$.

Ensure: Calibrated classifier, conformal thresholds, and $\tau^\star$.

1: Train f_θ on $\mathcal{D}_{\text{train}}$; obtain logits $z_\theta(x)$.

2: Fit temperature T on $\mathcal{D}_{\text{cal}}$ (NLL minimization); define $p_T(y \mid x)$.

3: **Choose conformal mode and compute thresholds:**

4: (i) *Vanilla*: compute q_α on $\{a_i = 1 - p_T(y_i \mid x_i)\}$.

5: (ii) *Mondrian*: for each group g, compute $q_{\alpha,g}$ on $\mathcal{D}^g_{\text{cal}}$.

6: (iii) *Weighted*: train logistic discriminator (calibration vs. target covariates) to obtain weights $w_i \propto r(x_i)$; compute weighted $\tilde{q}_\alpha$.

7: For each $\alpha \in \mathcal{A}$ and $\tau \in \mathcal{T}$, evaluate $\mathcal{L}_{\text{cal}}(\tau; \alpha)$ in (6); retain $(\alpha^\star, \tau^\star)$ with minimum cost.

8: Fix $(\alpha^\star, \tau^\star)$. For input x, form $\mathcal{C}_{\alpha^\star}(x)$ and $u(x) = \max_y p_T(y \mid x)$. If $u(x) < \tau^\star$, defer; otherwise output $\hat{y}(x) = \arg\max_y p_T(y \mid x)$.

9: Evaluate on $\mathcal{D}_{\text{id}}$ and $\mathcal{D}_{\text{ood}}$: report expected cost, risk–coverage, ECE, and safety metrics at clinically relevant sensitivity.

Algorithm 1. Conformal triage with cost-aware deferral.

Experimental setup and results

We evaluate the proposed Triage-CP framework on early sepsis prediction using the PhysioNet 2019 Sepsis Challenge dataset[34]. The experiments are designed to test three questions: whether risk-based deferral coupled with conformal prediction reduces error on retained cases at fixed coverage; whether the operating point selected on a held-out calibration split degrades gracefully under temporal distribution shift; and whether advanced conformal variants, namely Mondrian and weighted conformal prediction, improve subgroup coverage balance and out-of-distribution (OOD)[35] robustness without sacrificing efficiency.

Dataset and preprocessing

The PhysioNet 2019 cohort[34] (available at https://physionet.org/content/challenge-2019/1.0.0/) contains hourly time-series measurements for intensive care unit patients. Following standard practice, we cast sepsis detection

as a 6-hour early prediction task, selecting a reference hour six hours before the first positive sepsis label when available for positive cases, and an hour at least six hours after admission for negative cases. From each reference hour we extract a fixed-length feature vector of routinely collected variables (vital signs, common laboratory values, and related measurements). Missingness[36] is addressed by last-observation-carried-forward imputation along the patient trajectory, with any remaining gaps set to zero after standardization so that imputed values do not introduce scale artifacts. For fairness analysis with Mondrian conformal prediction we also extract patient age and gender as demographic covariates; these attributes are used only to define subgroups and are not part of the classifier input. After preprocessing, the analytic sample contains 2,155 patients with a sepsis prevalence of 7.0%.

To emulate prospective deployment, we partition the data into four non-overlapping splits. The training set contains 1,036 patients (48.1%), the calibration set 344 patients (16.0%), the in-distribution (ID) test set 344 patients (16.0%), and the OOD test set 431 patients (20.0%). The OOD cohort is constructed by temporal separation so that later-admitted patients form the test distribution, which introduces a mild but realistic shift in case mix and practice patterns. All model and threshold selection is performed without exposure to the test splits. For clarity, the empirical prevalences differ slightly across evaluation splits: 5.81% on the ID test split and 7.19% on the OOD test split.

Model architecture and training

The base classifier is a multilayer perceptron with two hidden layers of sizes 256 and 128, rectified linear activations, and dropout at rate 0.1. We train with AdamW using a learning rate of 10^{-3}, weight decay of 10^{-4}, batch size 256, and a fixed schedule of 10 epochs, which was sufficient to reach convergence in loss and stable calibration on the development splits. To obtain reliable probabilistic outputs, we apply temperature scaling on the calibration split. The scalar temperature is optimized to minimize the negative log-likelihood[37] and is then fixed for all subsequent evaluation.

Conformal prediction methods

We compare three constructions that share the same nonconformity score but differ in how calibration quantiles are computed. Vanilla split conformal prediction uses a single quantile on the calibration scores to target marginal coverage. Mondrian conformal prediction computes separate quantiles within gender-defined subgroups, which enables group-conditional coverage monitoring and fairness analysis. Weighted conformal prediction replaces the empirical quantile with an importance-weighted quantile to better match a shifted target distribution; we estimate density ratios by fitting a logistic regression classifier to discriminate calibration versus OOD covariates and transform its probabilities into weights. We assume access to unlabeled covariates from the OOD evaluation window for this weighting step. All methods are configured to target 90% coverage by setting $\alpha = 0.1$. For weighted conformal prediction, we assume access to unlabeled covariates from the target (OOD) time window, which are typically observable at deployment as model inputs; no labels from the test splits are used, and this step is solely for shift estimation rather than performance tuning.

Deferral policy and cost matrix

Deferral follows a certainty-threshold rule on the calibrated posterior, where a case is routed to human review if the maximum predicted class probability falls below a threshold τ. In keeping with clinical triage principles, we select τ to minimize an expected clinical cost on the calibration split using a grid search over $\{0.6, 0.7, 0.8, 0.85, 0.9, 0.95\}$. The cost function is asymmetric to reflect the higher harm of missed sepsis: a false negative incurs cost 10.0, a false positive cost 1.0, and a deferral cost 2.0. The same operating point is then applied unchanged to ID and OOD test sets.

Experimental results

Figure 2 summarizes the risk–coverage trade-off induced by the certainty-threshold deferral rule. For a threshold τ, we defer when $\delta_\tau(x) = 1\{u(x) < \tau\}$, where $u(x) = \max_y p_T(y \mid x)$ is the calibrated certainty, and we define coverage as $\kappa(\tau) = \mathbb{P}\{u(X) \geq \tau\}$, the fraction of cases for which the model issues a prediction (i.e., does not defer). Because coverage decreases monotonically as τ increases, the curve is plotted as risk versus coverage rather than versus τ, and the points appear ordered by $\kappa(\tau)$: larger thresholds correspond to the left side of the plot (lower coverage), while smaller thresholds correspond to the right side (higher coverage). Consequently, the smallest non-zero coverage shown reflects the most selective (largest) threshold used in the sweep, whereas $\tau = 0.6$ yields the highest coverage among the candidate thresholds and appears near the right edge.

As selectivity increases and coverage falls, risk decreases, confirming that the certainty score ranks cases by difficulty and concentrates errors among deferred cases. The same qualitative pattern holds under temporal shift, with only a modest gap between the ID and OOD curves. At the operating point corresponding to 80% coverage, the retained-case error is reduced by 49.6% on ID and 46.7% on OOD relative to predicting on all cases, indicating that uncertainty-based deferral remains informative even when the evaluation distribution differs from development.

Optimizing the deferral threshold on the calibration split selects $\tau = 0.8$. Evaluated at this point, the expected clinical cost is 0.5640 for the ID test set and 0.7053 for the OOD test set (Fig. 3). The relative increase of 25.1% on OOD quantifies the impact of temporal shift, but the absolute values remain within a regime that would be acceptable under monitored deployment. Crucially, the operating point is chosen to minimize an explicit harm proxy rather than to maximize accuracy, which aligns the system with safety-first objectives.

Probability calibration underpins both conformal set construction and the defer rule. After temperature scaling, expected calibration error is 0.016 on ID and 0.030 on OOD (Fig. 4). Although the relative increase OOD is substantial, the absolute ECE remains below 0.03, which supports the use of calibrated posteriors to

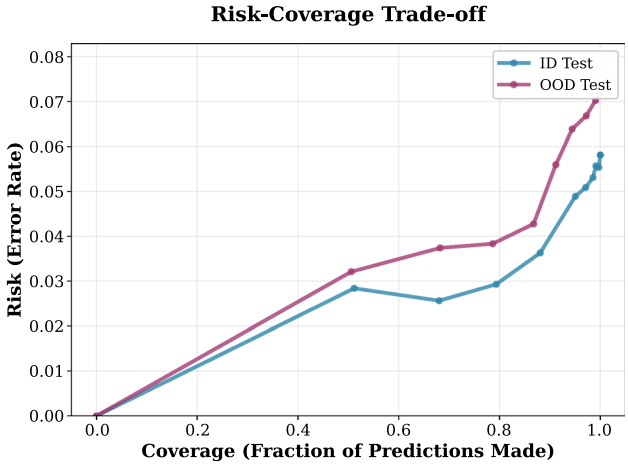


Fig. 2. Risk–coverage curves for the ID and OOD test sets. Coverage is defined as the fraction of cases on which the model issues a prediction (i.e., non-deferred cases) under the certainty-threshold rule $\delta_\tau(x) = 1\{u(x) < \tau\}$ with $u(x) = \max_y p_T(y \mid x)$, so larger τ corresponds to lower coverage. The curve is plotted as risk versus coverage (not as a function of τ), and points are therefore ordered by coverage: the left side corresponds to more selective (larger) thresholds and the right side to less selective (smaller) thresholds. At 80% coverage, risk drops by 49.6% (ID) and 46.7% (OOD) compared to making predictions on all cases.

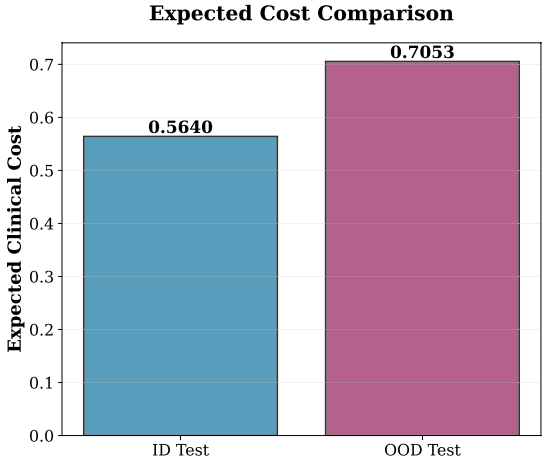


Fig. 3. Expected clinical cost comparison. The optimal deferral policy $\tau = 0.8$ achieves low expected cost on both ID (0.5640) and OOD (0.7053) test sets. The approximately 25% cost increase on OOD data is moderate, indicating reasonable robustness to temporal distribution shift.

rank cases by confidence and to form nonconformity scores for conformal prediction. This level of calibration is helpful for clinical interpretability because reported confidence is commensurate with empirical frequency. The resulting specificity is intentionally low, reflecting a safety-first policy that prioritizes missed-case avoidance and permits deferral.

At a sensitivity target of 95%, chosen to reflect conservative clinical practice, negative predictive value is approximately 1.0 on both ID and OOD test sets (Fig. 5). This suggests that when the system rules out sepsis, the decision is highly reliable, which directly serves the goal of avoiding missed diagnoses. As expected for a safety-oriented operating point, specificity is lower, reflecting a deliberate tilt toward sensitivity and deferral over aggressive rule-in.

Distribution shift affects both discrimination and calibration. Fig. 6 summarizes the magnitude of change by reporting relative increases in expected cost and calibration error. While ECE nearly doubles in relative terms, the absolute ECE remains low. The cost increase is moderate rather than severe, which, together with stable risk–coverage curves, supports the view that a calibration-selected operating point can be carried forward to deployment provided that performance is monitored and periodic recalibration is feasible.

All three conformal constructions achieve coverage close to the nominal 90% target on the ID test set, which empirically corroborates their marginal validity under exchangeability (Fig. 7). On the OOD cohort, coverage declines modestly for all methods, with Weighted CP exhibiting the smallest degradation (88.2%) and Mondrian CP also improving upon Vanilla Split CP. The small absolute changes indicate that conformal sets

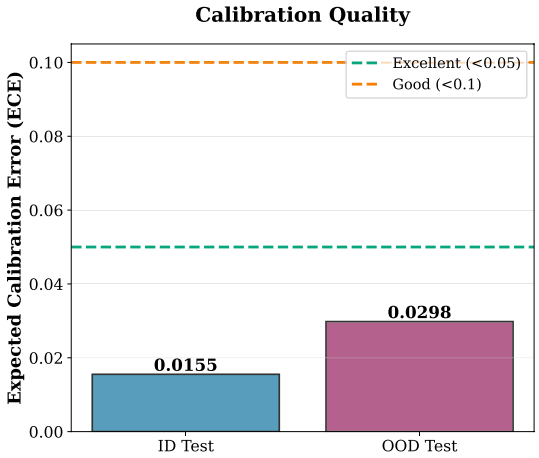


Fig. 4. Expected calibration error (ECE) for ID and OOD test sets. Temperature scaling achieves excellent calibration on both distributions, with ECE below 0.03. Reference lines indicate qualitative thresholds at 0.05 and 0.1. Well-calibrated predictions are essential for trustworthy uncertainty quantification in clinical settings.

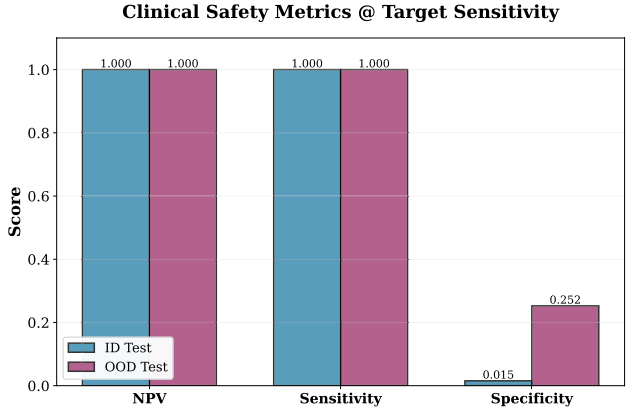


Fig. 5. Clinical safety metrics at target sensitivity 95%. The system attains near-perfect negative predictive value on both ID and OOD test sets, indicating that when ruling out sepsis, the model is almost always correct. High sensitivity ensures minimal missed sepsis cases, with an expected trade-off in specificity.

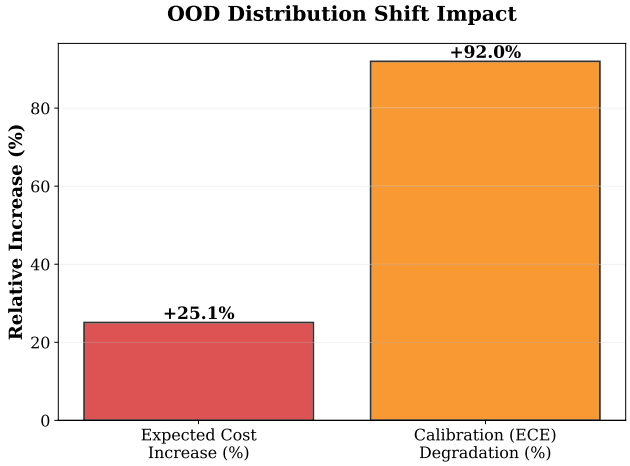


Fig. 6. Impact of distribution shift on model performance. OOD degradation is summarized by the relative increases in expected cost and calibration error. While calibration degrades in relative terms, the absolute ECE remains low (0.030). The moderate cost increase indicates acceptable robustness with temporal monitoring.

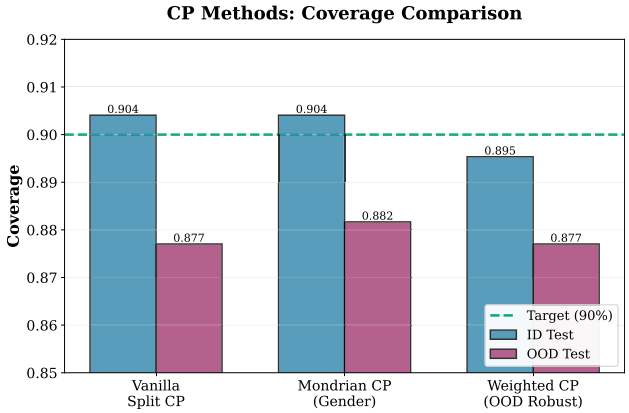


Fig. 7. Coverage comparison across conformal methods. All methods reach coverage close to the 90% target on ID data. Under OOD shift, coverage drops modestly: Weighted CP achieves 88.2% (1.8pp drop), Mondrian CP 87.8% (2.2pp), and Vanilla Split CP 87.3% (2.7pp), confirming the robustness benefit of importance weighting and the additional value of stratification.

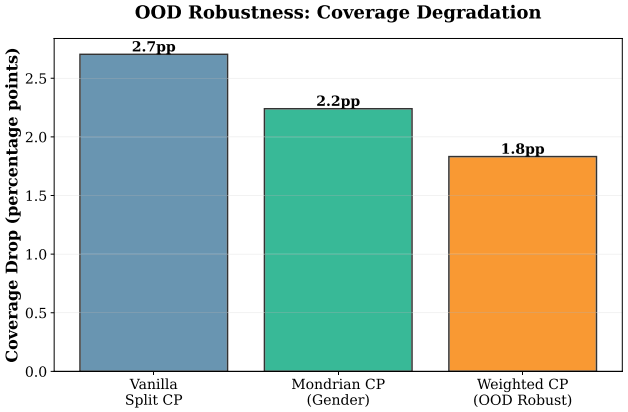


Fig. 8. Coverage degradation under distribution shift. Robustness is measured by the coverage drop from ID to OOD. Weighted CP shows the smallest drop (1.8pp) by aligning calibration scores to the target distribution via importance weighting; Mondrian CP also reduces degradation relative to Vanilla Split CP (2.2pp vs. 2.7pp).

remain informative even when the case mix shifts over time. In addition to overall coverage and robustness, we assess subgroup coverage to support fairness-oriented auditing. The gender-stratified Mondrian conformal results are reported in Section 5.

Weighted conformal prediction mitigates coverage loss under covariate shift by reweighting calibration scores toward the target distribution, which reduces the discrepancy between the empirical calibration quantile and the desired target quantile (Fig. 8). Mondrian conformal prediction, although motivated by subgroup validity, also exhibits improved robustness relative to Vanilla Split CP, suggesting that stratification can curb some forms of shift when the grouping variable correlates with the changing case mix.

A potential concern with set-valued prediction is that prediction sets could become large and uninformative. In practice, the average set size remains near one across methods (Fig. 9), which means that decisions typically reduce to a single class, while the occasional larger set flags uncertainty and invites deferral. Importantly, the gains in subgroup balance and OOD robustness from Mondrian and Weighted CP do not come at the cost of inflated set sizes, preserving specificity in guidance to clinicians.

The results show that Triage-CP delivers an attractive operating profile for safety-critical early sepsis prediction: calibrated posteriors and conformal sets provide transparent uncertainty; a single cost-minimizing deferral threshold yields a favorable risk–coverage trade-off and low expected clinical cost; and fairness- and shift-aware conformal variants improve coverage behavior without compromising efficiency. These properties are maintained under temporal shift, supporting the practical use of conformal triage with routine monitoring and periodic recalibration.

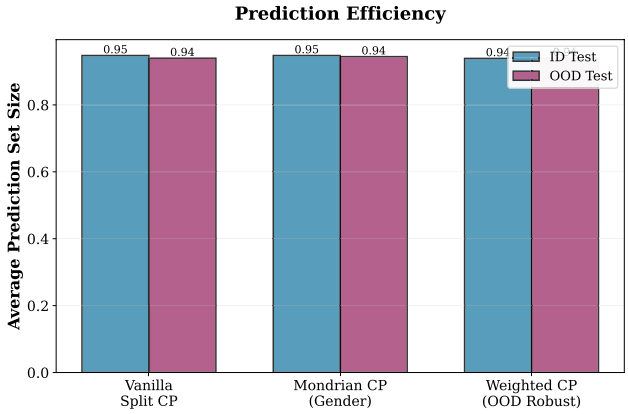


Fig. 9. Prediction set efficiency in a binary task. The vast majority of cases yield singleton sets; the average set size is close to 1 (median 1; 95th percentile 2), indicating that ambiguity is rare and naturally flags cases for deferral while preserving specificity across methods.

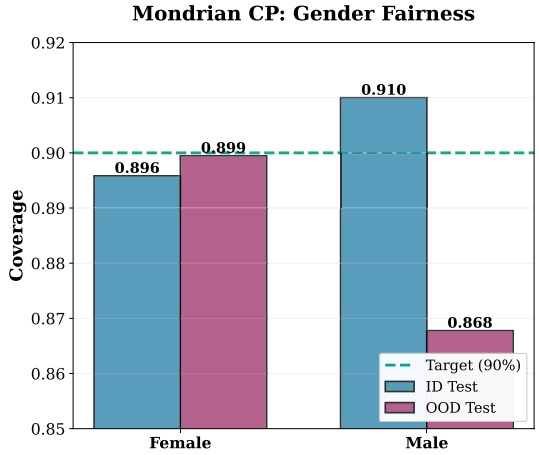


Fig. 10. Gender-stratified coverage with Mondrian conformal prediction. Computing separate quantiles for female and male patients produces balanced coverage. On ID data, the coverage gap is 1.4 percentage points (female 89.6% versus male 91.0%), with closely matched thresholds (female $q = 0.0576$, male $q = 0.0581$).

Discussion

Clinical interpretation and decision-theoretic perspective

The proposed framework reorients clinical prediction from leaderboard discrimination toward operational safety. Rather than committing to a label for every case, the system admits uncertainty and routes low-confidence encounters to human review. This behavior is not an incidental by-product of threshold tuning but the explicit solution of a decision problem in which false negatives are far more harmful than false positives and in which deferral bears a finite, accountable cost. The empirical risk–coverage curves in Section 4 show that confidence-based selectivity reliably concentrates errors among deferred cases while preserving low error on retained cases. In practice, this offers a dial that a service can tune to match local capacity and risk tolerance.

Assumptions and validity conditions

The guarantees inherited from conformal prediction require exchangeability between calibration and target data, and the temperature-scaling step assumes that a single scalar can correct the calibration of the base network on the calibration split. Our temporal OOD construction intentionally breaks exchangeability to probe robustness; coverage degrades modestly in this setting, and the weighted variant partially recovers target behavior. These observations align with theory: marginal coverage is exact under exchangeability and approximate under shift, with the degree of approximation driven by how well the weighting scheme recovers the test distribution. Missingness handling via last-observation-carried-forward is simple and stable but shifts uncertainty to preprocessing; Mondrian CP presumes observed, relevant group labels and adequate subgroup size.

Fairness auditing via subgroup coverage

Group-conditional calibration is a practical requirement in many clinical settings. By computing gender-specific quantiles on the calibration split, Mondrian conformal prediction produces nearly balanced coverage on the ID

Split	Error rate (%)	AUROC	AUPRC
In-distribution (model)	5.8	0.6978	0.1657
In-distribution (always negative)	5.8	0.5000	0.0580
Out-of-distribution (model)	7.2	0.7904	0.2081
Out-of-distribution (always negative)	7.2	0.5000	0.0720

Table 1. Base model performance at 100% coverage (no deferral). For comparison, the trivial baseline always predicts the majority class (no sepsis), yielding AUROC 0.5 and AUPRC equal to the positive prevalence.

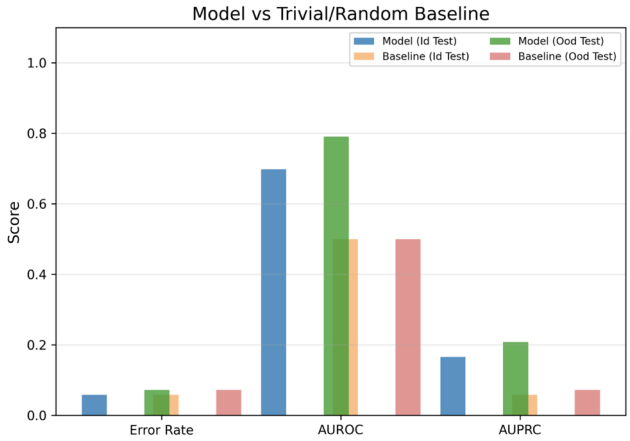


Fig. 11. Model performance versus trivial/random baselines at 100% coverage on in-distribution and out-of-distribution splits. While error rate is similar to the majority-class predictor due to class imbalance, AUROC and AUPRC demonstrate meaningful discrimination relative to random performance.

test set, with only a 1.4 percentage point difference between female and male subgroups (Fig. 10). The learned thresholds are very close, consistent with small differences in subgroup prevalence. This behavior illustrates how conformal methods can support fairness auditing by reporting both overall and subgroup coverage without altering the underlying classifier.

Base model performance at full coverage

Because sepsis prevalence is low, the overall error rate at 100% coverage can be dominated by true negatives, and a trivial majority-class rule can appear competitive while providing no clinically useful discrimination (zero sensitivity by construction). We therefore report standard threshold-free discrimination metrics for the base classifier at full coverage, alongside the corresponding ROC and precision–recall curves. As summarized in Table 1, the base model attains AUROC of 0.6978 on the in-distribution test split and 0.7904 under temporal shift, indicating non-trivial ranking ability. The area under the precision–recall curve (AUPRC) is 0.1657 (in-distribution) and 0.2081 (out-of-distribution), which is substantially above the random baseline given by the positive prevalence (0.058 and 0.072, respectively). Figure 11 contrasts these metrics against trivial/random baselines, and Fig. 12 visualizes the corresponding ROC and precision–recall curves. These results support that the base model has meaningful discriminative power despite class imbalance, and that its confidence scores provide an informative signal for the subsequent selective prediction and deferral analyses.

Cost function dependence, landscape visualization, and sensitivity analysis

Figure 13 visualizes the expected-cost landscape over a dense grid of τ values for both ID and OOD settings, shown on calibration and test splits alongside the trivial baseline. This plot makes clear that the objective can remain relatively flat across a range of operating points and then increase sharply as τ grows, consistent with increasing deferral and its associated penalty. To connect the cost trajectory to operational behavior, Fig. 15 additionally reports the test-set expected cost together with coverage as a function of τ, highlighting the trade-off between deferring more cases and reducing error-driven cost terms. Using a fine-grained sweep, the minimum test-set expected cost occurs at $\tau = 0.84$ on ID and $\tau = 0.81$ on OOD.

The results further indicate that, under the original cost setting, performance relative to the trivial baseline can differ between evaluation conditions: the optimized operating point can be slightly worse than the trivial always-negative baseline on the ID split while improving over the trivial baseline on the OOD split. This behavior is consistent with limited discriminative signal under heavy class imbalance, where small changes in deferral and errors can offset each other in the expected-cost objective. To quantify how conclusions depend on the cost specification, we conduct a sensitivity analysis across alternative penalty ratios for false negatives, false positives, and deferral. Figure 14 compares optimized expected cost against the trivial baseline across configurations for both ID and OOD test sets, and Fig. 16 reports the corresponding optimal thresholds, showing that the minimizer

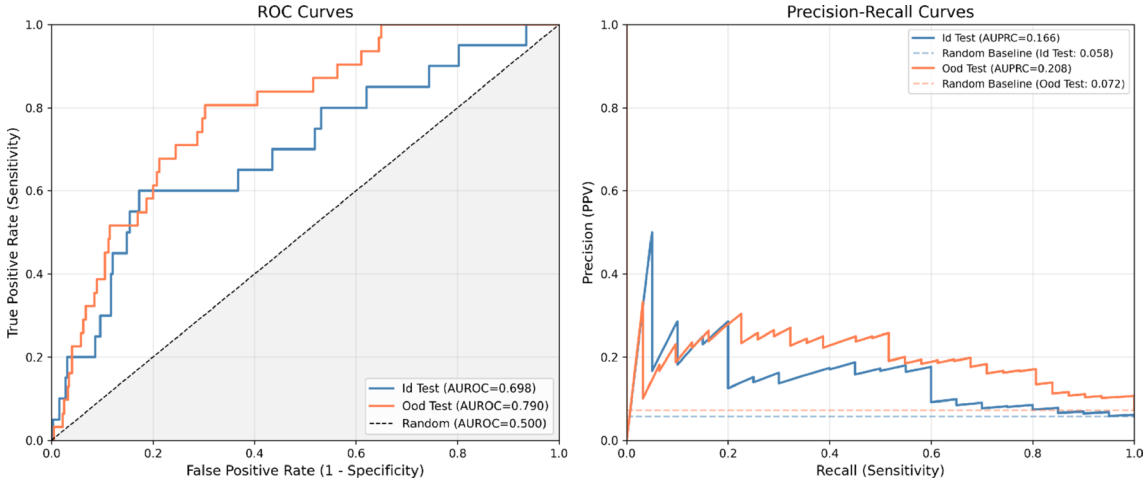


Fig. 12. ROC (left) and precision–recall (right) curves for the base model at 100% coverage. The dashed references correspond to random performance (AUROC 0.5) and the prevalence baselines for AUPRC (in-distribution 0.058; out-of-distribution 0.072).

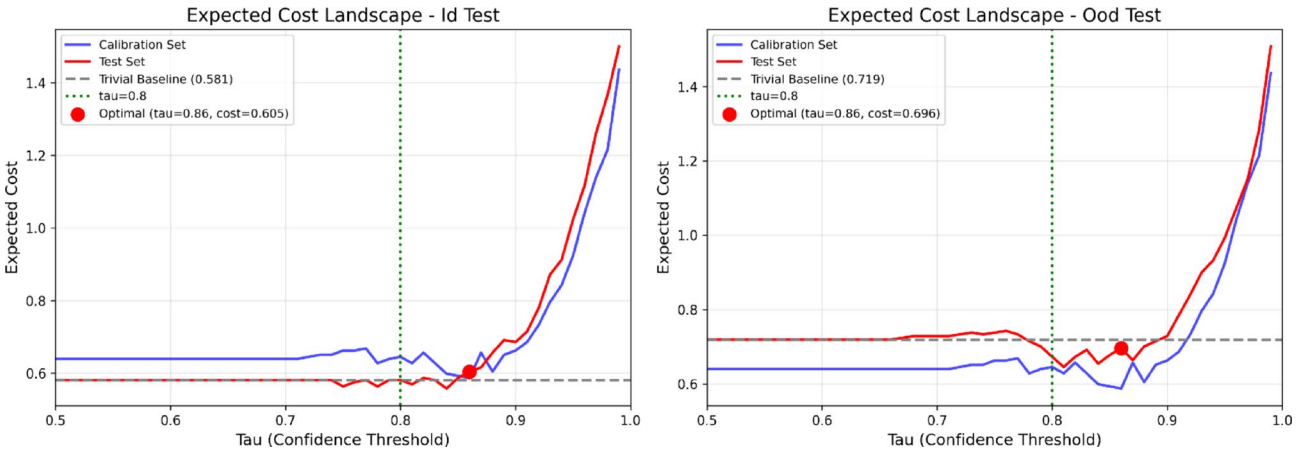


Fig. 13. Expected cost as a function of the confidence threshold τ on ID and OOD settings, shown for calibration and test splits. The dashed horizontal line denotes the trivial always-negative baseline cost (prevalence $\times$ c_{FN}). The vertical line indicates $\tau = 0.8$, and the marker indicates the calibration-selected optimum under the original costs.

of expected cost can shift substantially as the weights change. Figure 17 summarizes the relative improvement over the trivial baseline by configuration and setting, demonstrating that improvements can change in both magnitude and direction as the cost weights vary, as expected in cost-sensitive selective prediction. In particular, meaningful improvements emerge when false negatives are penalized more strongly and/or deferral is relatively less costly, including High FN (20 : 1 : 2), Low Defer (10 : 1 : 1), and Clinical High (50 : 1 : 5), where improvements reach approximately 18.5% on ID and up to 30.0% on OOD relative to the trivial baseline (Fig. 17). These results emphasize that the cost weights are not treated as universally valid clinical utilities; instead, the framework is intended to be instantiated with site- and task-specific utilities, and the sensitivity analysis is provided to make this dependency explicit.

Choice of cost model and operating point

The operating point emerges from minimizing an explicit expected cost on the calibration split. This cost encodes institutional preferences and regulatory constraints more faithfully than discrimination metrics and exposes trade-offs that clinicians already reason about informally. Because the policy is post hoc and calibrated, the same trained model can be re-optimized for a new cost profile without re-fitting, which is advantageous for multi-site deployment. In practice, a coarse grid suffices to expose the cost curve; a narrower band around the minimizer can be explored prospectively to refine $\tau^{\star}$ while guarding against overfitting to small calibration sets.

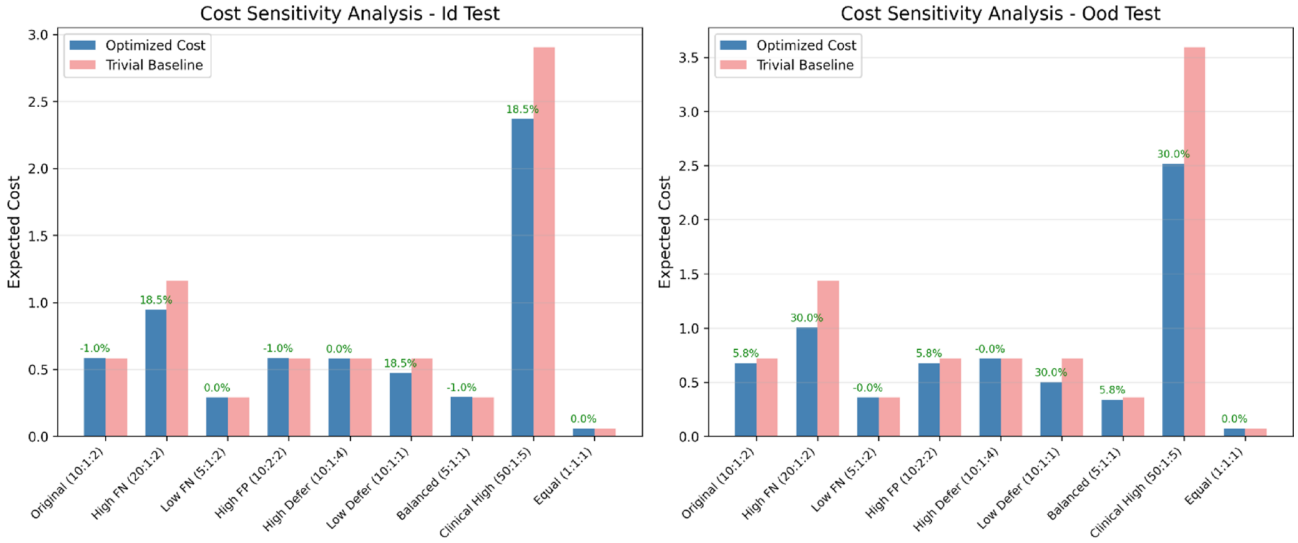


Fig. 14. Sensitivity analysis across alternative cost configurations. Bars compare the optimized policy cost against the trivial always-negative baseline for both ID and OOD test sets, illustrating how improvements depend on the relative penalties for false negatives, false positives, and deferral.

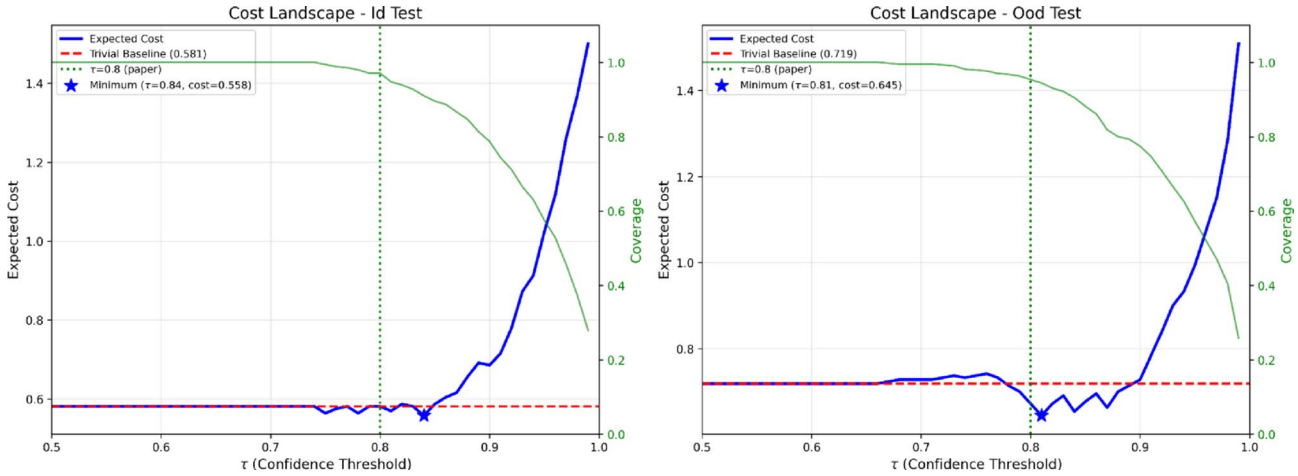


Fig. 15. Test-set expected cost landscape with coverage as a function of τ for ID and OOD settings. The minimum of the cost curve and the reference $\tau = 0.8$ are shown to assess robustness of the operating point.

Robustness under distribution shift

Temporal separation between development and evaluation cohorts introduces measurable but controlled degradation in calibration and cost. Weighted CP mitigates this effect by reweighting calibration scores toward the target covariate distribution; its success depends on the quality of the density-ratio model. Routine recalibration and drift monitoring can further stabilize performance. The empirical stability of the risk–coverage trade-off across ID and OOD cohorts suggests that confidence-based deferral remains a reliable safety valve as case mix shifts.

Operationalization and governance

For deployment, transparency, tunability, and monitoring matter most. The pipeline exposes levers that a service can govern: the conformal level, the deferral threshold, and the cost vector. Because the operating point is determined on a held-out split, changes can be reviewed and approved without altering the trained model. Dashboards should track coverage, deferral, expected cost, calibration error, and sensitivity at fixed rule-out thresholds, with alerts on drift. Deferral consumes clinician time; coupling thresholds to real-time capacity prevents safety gains from eroding access.

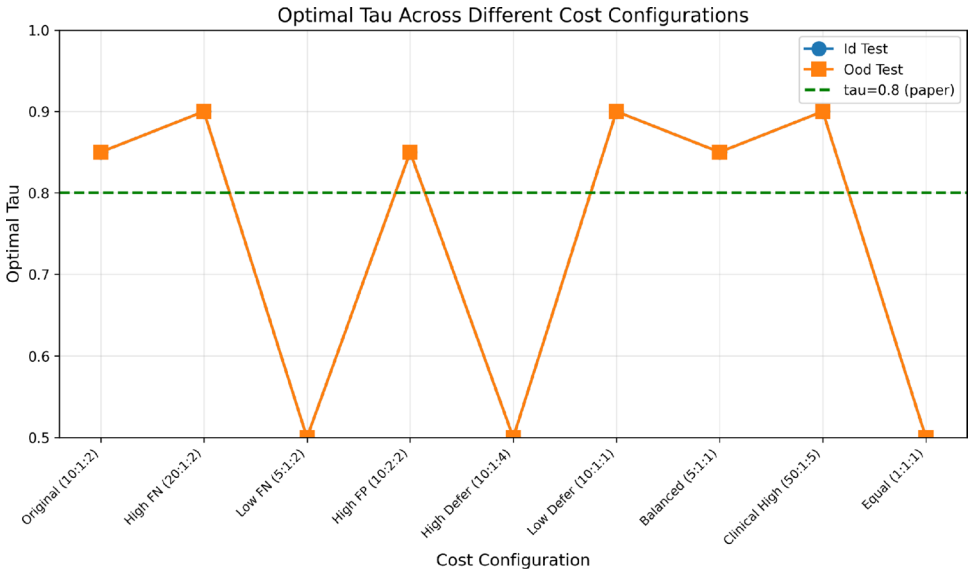


Fig. 16. Optimal τ selected under different cost configurations on ID and OOD test sets, with $\tau = 0.8$ shown as a reference.

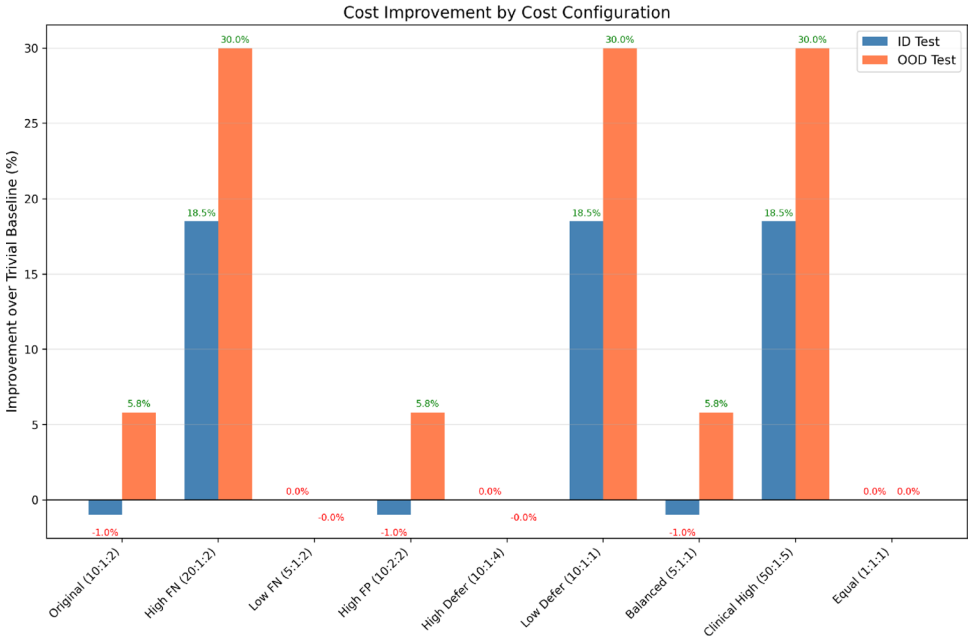


Fig. 17. Relative improvement in expected cost over the trivial always-negative baseline for each cost configuration, reported separately for ID and OOD test sets.

Ethical and practical considerations

Selective prediction redistributes workload between automation and clinicians. Excessive deferral can burden staff or delay care if escalation pathways are not resourced. Clear communication of uncertainty and rationale for deferral, routine subgroup analyses, and logged change management are essential. The public-data implementation lowers the barrier to replication but is not a substitute for site-specific validation.

Summary of limitations

To support transparent interpretation and deployment planning, Table 2 summarizes the main limitations of this study, their practical implications, and plausible mitigation strategies.

Limitation	Practical implication	Mitigation and future work
Single public cohort and a tabular base model	External generalization across institutions, modalities, and workflows is not established.	Evaluate multi-site cohorts and alternative base learners, and report external validation.
Surrogate cost model for harm and workload	The selected operating point depends on assumed relative costs and local capacity.	Perform sensitivity analysis over plausible cost ranges and align costs with workflow constraints.
Coverage guarantees rely on exchangeability	Conformal coverage can degrade when distribution shift violates exchangeability.	Monitor coverage under shift, update calibration as needed, and consider shift-aware conformal variants.
Importance-weighted conformal prediction assumptions	The method requires unlabeled target covariates and accurate density-ratio estimation.	Assess robustness to imperfect weights and compare alternative weighting or adaptation strategies.
Limited subgroup auditing scope and preprocessing dependence	Coverage parity for predefined groups does not fully capture equity and can depend on preprocessing.	Extend subgroup audits and test preprocessing variants to check stability of conclusions.

Table 2. Summary of limitations, implications, and mitigation strategies.

Item	Summary
Framework	A calibrated probabilistic classifier is combined with conformal prediction sets and a cost-aware deferral rule that routes low-confidence cases to clinician review.
Risk–coverage benefit	At 80% coverage, the error rate on retained cases decreases by 49.6% in-distribution and 46.7% under temporal shift.
Cost robustness under shift	A single deferral threshold chosen on the calibration split attains low expected cost on both test splits, with a moderate increase under temporal shift.
Calibration and conservative rule-out	Expected calibration error remains low (0.016 in-distribution; 0.030 out-of-distribution), and rule-out performance is conservative at a 95% sensitivity target.
Coverage and auditing	Coverage stays near the nominal 90% target in-distribution with only slight degradation out-of-distribution, and subgroup coverage auditing shows a small gender gap.

Table 3. Compact summary of the framework and main empirical findings on in-distribution and out-of-distribution test splits.

Conclusion

This paper presented a selective prediction framework for clinical triage that unifies calibrated probabilistic modeling, conformal set construction, and a cost-aware deferral rule. On the PhysioNet 2019 early sepsis task, the approach achieved a favorable operating profile that is aligned with safety priorities rather than leaderboard accuracy. By optimizing a single certainty threshold on a calibration split, the system reduced retained-case error by 49.6% on in-distribution data and 46.7% under temporal shift at 80% coverage, and it minimized an explicit clinical harm proxy with expected costs of 0.5640 (ID) and 0.7053 (OOD) at the chosen operating point. Temperature scaling delivered well-calibrated probabilities (ECE 0.016 ID; 0.030 OOD), supporting trustworthy uncertainty quantification and stable conformal sets. At a clinically relevant sensitivity target of 95%, negative predictive value was approximately 1.0 on both splits, demonstrating safe rule-out behavior with the expected specificity trade-off. Advanced conformal variants preserved efficiency while adding desirable properties: coverage remained close to the 90% target ID with only modest OOD degradation, Weighted CP yielded the smallest coverage drop under shift, and Mondrian CP narrowed the gender coverage gap to 1.4 percentage points. Together, these results indicate that conformal guarantees coupled with cost-aware deferral produce transparent, tunable policies that degrade gracefully when the deployment distribution changes. Table 3 provides a compact overview of the proposed pipeline and the principal empirical outcomes to support quick reading and comparison.

Future work will extend the framework along methodological, operational, and clinical dimensions. Methodologically, richer calibration families[38] and conditional[39] or distributionally robust conformal constructions[40] could further stabilize guarantees under shift, while joint optimization[41] of subgroup-aware costs would align operating points with equity objectives beyond coverage parity. Operationally, integrating the deferral policy with real-time capacity management and drift monitoring would enable adaptive thresholds that respect service constraints without retraining, and sequential variants could coordinate multi-stage pathways that combine rapid rule-out with confirmatory testing. Clinically, broader evaluation across institutions, modalities, and base learners, together with prospective decision-curve analyses and user studies, will be needed to quantify workflow impact and acceptability. The public implementation accompanying this study is intended to accelerate such investigations and to support reproducible, safety-first deployment of selective prediction in medicine.

Data availability

This study used publicly available, de-identified data from the PhysioNet/Computing in Cardiology Challenge 2019, accessible at https://physionet.org/content/challenge-2019/1.0.0/. All custom code and scripts are archived on Zenodo at https://doi.org/10.5281/zenodo.18621127.

Code availability

All custom code and scripts used to generate the results reported in this study are publicly available in a DOI-minting repository. The code is archived on Zenodo at https://doi.org/10.5281/zenodo.18621127. The

corresponding source repository is hosted at https://github.com/c15926/conformal-triage-public__.

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Author contributions
Hyun Kwon: conceptualization, methodology, writing—original draft, software, validation, formal analysis, visualization. Dae-Jin Kim: writing—original draft, visualization, supervision.

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Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
This study used exclusively publicly available, de-identified data from the PhysioNet/Computing in Cardiology Challenge 2019[34]. As the dataset is fully anonymized and openly accessible under the PhysioNet Credentialed Health Data License, no ethical approval or informed consent was required. There are no human or animal experiments in this paper.

Additional information
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